Sensor less Speed Estimation of a Brushed DC Motor Using Feedback Current Ripple Counting

Practical Project as elective course

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ABSTRACT

This project investigates the application of a current ripple counting technique for sensorless speed estimation in a brushless DC (BLDC) motor. Reducing system cost and design complexity is a key objective in motor control applications, where conventional speed and position sensing methods rely on additional hardware such as optical encoders, Hall-effect sensors, and current transformers. While effective, these sensors increase component count, cost, and system complexity.

To eliminate the need for physical sensors, this work employs a sensorless motor control approach based on the analysis of motor phase current feedback. The inherent current ripples produced during the commutation process of the BLDC motor are utilized as a source of speed information. The sensed current signal is processed using a Butterworth low-pass filter to suppress noise and high-frequency distortions while preserving the fundamental ripple characteristics. A Schmitt trigger is then applied to convert the filtered analog signal into a clean digital pulse train suitable for reliable counting.

Motor speed is estimated by measuring the time period and counting the number of pulses generated by the commutation-related current ripples. Experimental data were collected using a current-sensing printed circuit board, a propeller-driven BLDC motor, and an oscilloscope. The acquired signals were further analyzed and processed in MATLAB. The proposed method demonstrates accurate and robust speed estimation over a frequency range of 20 Hz to 120 Hz, confirming the effectiveness of ripple-based sensing across varying motor speeds.

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1. Introduction

Brushed DC motors are used extensively in a huge range of consumer and industrial products, such as pumps, air fans and automotive window winders. These products often benefit from some form of speed control, but the addition of speed measurement hardware would negatively impact on costs. First, there is the cost of the sensor itself, which is commonly a tachometer or position sensor. Next, there is the mounting cost which includes the extra space and housing required to contain the sensor [3]. Other costs include connectors, cabling and sensor conditioning with both electronics and computation costs. Reliability of sensors may also be an issue in non-office environments where dust, pollution and vibration can all affect reliability. Clearly, there is a need for 'sensorless' speed measurement which requires no extra hardware. While sensorless speed measurement is most applicable to small motors in cost sensitive applications, even large motors can use these methods as a backup in case of a fault in the hardware speed sensor.

The sensorless motor control is a type of drive technique that does not rely on sensors to determine the actual motor position or speed. It determines the position of the motor based on electrical parameters such as Back-EMF or motor current feedback, which is present during the motor run-time. The current feedback is used for circuit protection, motor speed estimation or position control. The motor current can be monitored by placing a current sense resistor in a series with the motor. Using a current sense amplifier tapped on the resistor terminals, the current detected is converted to a measurable voltage that acts as a position or speed feedback signal to a closed-loop motor control. In this application, the periodic variations in current feedback, due to BDC motor electro-mechanical construction, are observed. The displacement taken by the motor rotation can be identified by counting the number of those periodic variations during motor operation [1]. The method of using motor current feedback to determine motor position and speed is referred to as the ripple counting technique and will be discussed and utilized in this application note. An electric motor mainly consists of a rotor and a stator and can be either brushed or brushless. The rotor is the part that is rotating, and the stator often sits around the rotor. The motor can be driven by either direct current (DC) or alternating current (AC). There are some major differences in how the motor works depending on if it is brushed or brushless. In a brushed DC motor, the stator mainly holds field magnets, and the rotor, wire windings. To make the rotor rotate, a commutator is used.

The commutator sits on the shaft of the rotor and is used to reverse the polarity of the rotor windings so that the motor keeps rotating. When the motor is rotating because of the attraction between field magnets and rotor windings, the commutator, which is made out of conducting materials, loses its connection with the brushes[5]. The brushes later get contact with the commutator and the polarity is switched. Brushed DC motors are easily constructed and does not need any computers since the hardware handles the commutations. The speed of the motor is controlled by changing the supply voltage. One problem with the brushed DC motor is that it needs maintenance, since the brushes are worn out over time because of the contact with the commutator.

1.1 Brushed DC Motor current Ripple

Ripple Counting works by detecting current ripples that occur during commutation of the armature windings in a BDC motor. Commutation is the process of current reversal in the windings of a BDC motor to make sure that the motor torque remains in the same direction during motor operation. The contact brushes used with the commutator cause momentary short circuits in the motor windings which appear as ripples in the observed motor current waveform.

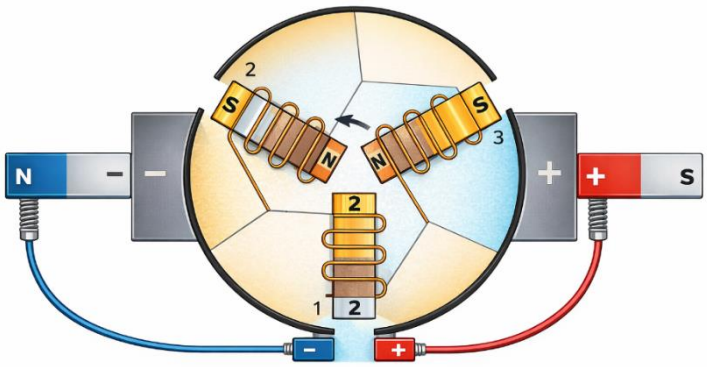


Figure 1 : DC Motor construction with 2 brushes and 3 commutators

Referring to the Figure1 might be useful to follow the rest of this discussion [5] .

Measuring the ripples per unit time provides the ripple speed (or ripple frequency), w_R (in Hz). To convert into rad/s, use the following equation:

$w_R(in\ rad/s)$= w_R (in Hz) × 2π (1)

Knowing the number of commutator segments and brushes, the motor speed, w_M, can then be calculated by dividing the ripple frequency by the number of ripples per revolution. Number of ripples per revolution, N_R, can be found by taking the lowest common multiple (LCM) of the number of brushes, N_B, and the number of commutator segments, N_C.

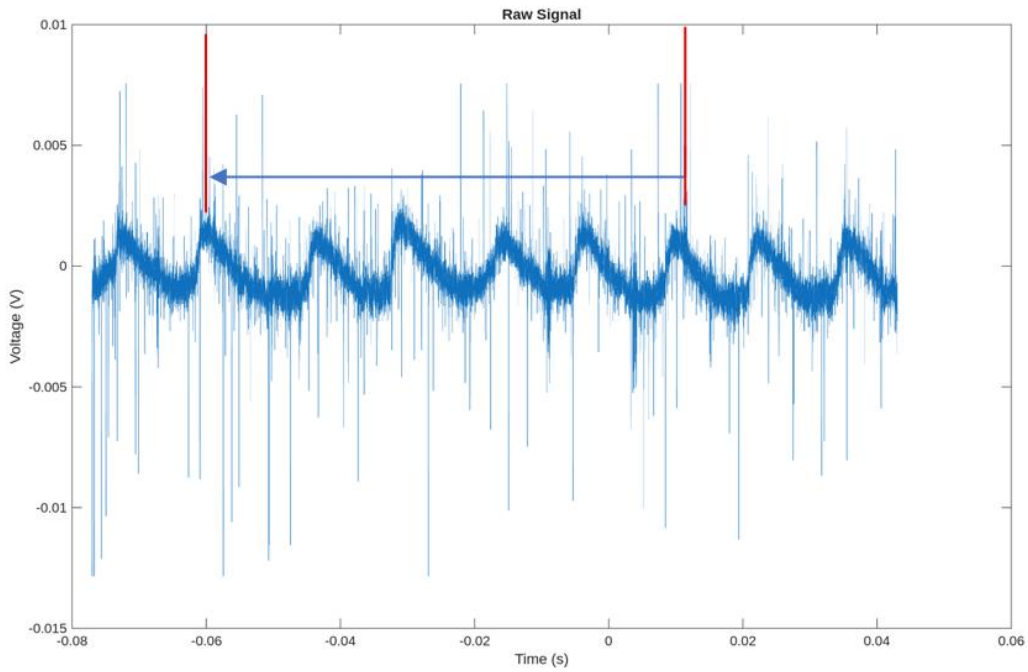
N_R =LCM (N_B, N_C) (2)

Hence, speed of the motor is given by:

$w_M(in\ rad/s)=\frac{w_R(in\ rad/s)}{N_R}$ (3)

According to the specification of implemented DC Motor, we have NB =2, and NC=3. Hence, NR = LCM (2,3) = 6. This is verified by logic, since each brush short-circuits each pair of commutator segments once during each revolution. There are 2 brushes and 3 commutators causing 6 short circuits per revolution. As a result, there are 6 ripples per revolution. Figure 2 shows an example current waveform during steady state operation. As the armature of a DC motor rotates, an electrical voltage spike is generated every time a brush moves from one

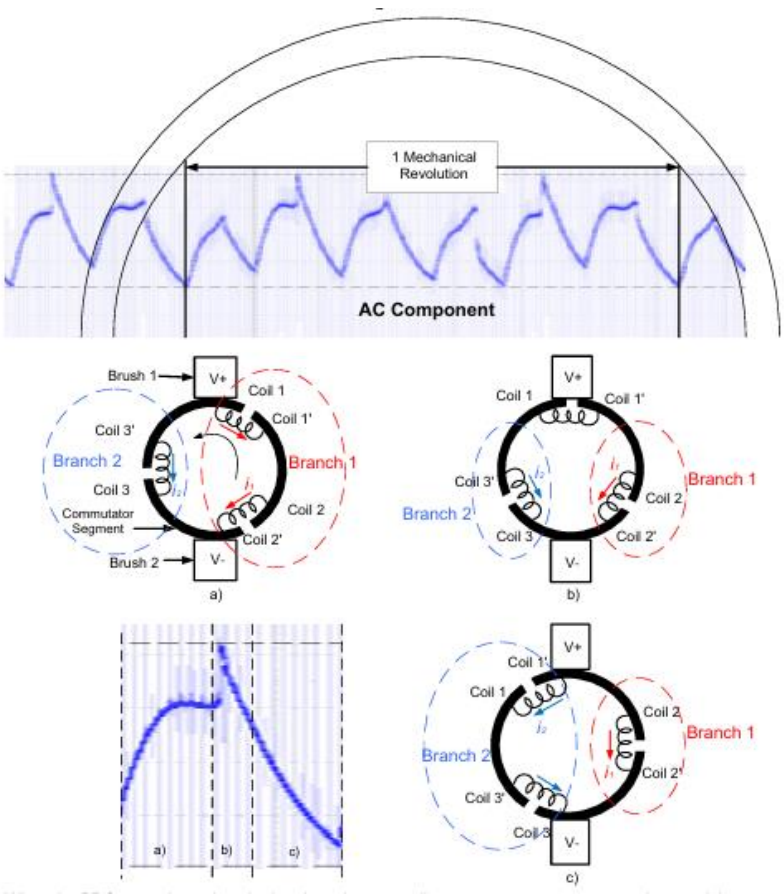
commutator segment to the following. It's called commutation spike and can be used to determine the speed that the motor is rotating at without an encoder.



2 Figure Brushed DC Motor Current Waveform with Ripples

1.2 Commutation and Current Ripple in the Rotating Motor

There are several types of ripples in a spinning motor. They can be classified as torque ripple, speed ripple and current ripple. The torque ripple is characterized by undesirable variation of torque production during shaft revolution, preventing smooth motor rotation and is observable mostly in Low-Frequency mode. The speed ripples, also known as speed fluctuations, are induced by parasitic torque pulsations that vary periodically with rotor position [4]. The current ripple is produced during motor rotation and defined as the low-amplitude current alternation, riding on the DC voltage supply. This current ripple is a periodic variation in current created by the rotor movement when it connects and disconnects rotor coils to the power supply via the brushes. The produced current ripple gives an insight about the commutation process happening in a BDC motor. Figure 3 shows how a three-commutator segment motor rotates and how the rotation produces current ripples or commutator spikes.



3 Figure: Ripple Production from a Rotating Motor

2. Butterworth Filter Design

The Butterworth Filter Design involves creating a maximally flat frequency response with no ripple in the passband or the stopband with a steep roll-off rate of -20dB per pole.

The Butterworth filter is an analogue filter design which produces the best output response with no ripple in the pass band or the stop band resulting in a maximally flat filter response but at the expense of a relatively wide transition band. In applications that use filters to shape the frequency spectrum of a signal such as in communications or control systems, the shape or width of the roll-off also called the "transition band [4].

2.1 Butterworth Filter Approximations

So far, we have looked at a low pass first circuits, their resultant frequency and phase responses. An ideal filter would give us specifications of maximum pass band gain and flatness, minimum stop band attenuation and also a very steep pass band to stop band roll-off (the transition band) and it is therefore apparent that a large number of network responses would satisfy these requirements. Not surprisingly then that there are a number of "approximation functions" in linear analogue filter design that use a mathematical approach to best approximate the transfer function we require for the filters design.

2.2 Low Pass Butterworth Filter Design

The frequency response of the Butterworth Filter approximation function is also often referred to as "maximally flat" (no ripples) response because the pass band is designed to have a frequency response which is as flat as mathematically possible from 0Hz (DC) until the cut-off frequency at -3dB with no ripples.

Higher frequencies beyond the cut-off point rolls-off down to zero in the stop band at 20dB/decade. This is because it has a "quality factor", "Q" of just 0.707. However, one main disadvantage of the Butterworth filter is that it achieves this pass band flatness at the expense of a wide transition band as the filter changes from the pass band to the stop band. It also has poor phase characteristics as well. The ideal frequency response, referred to as a "brick wall" filter, and the standard Butterworth approximations, for different filter orders are given below [6].

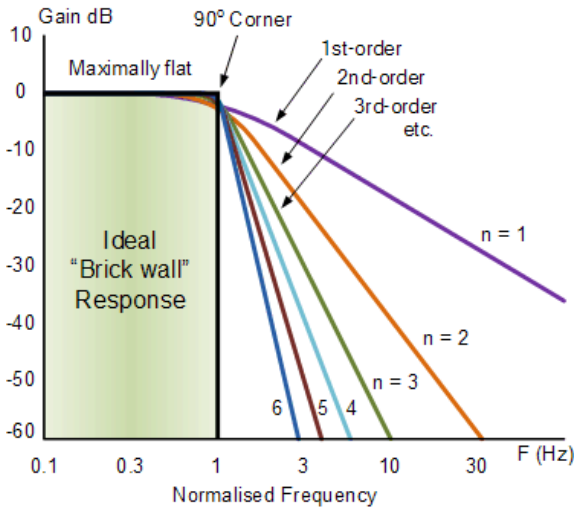


Figure 4: Ideal Frequency Response for a Butterworth Filter

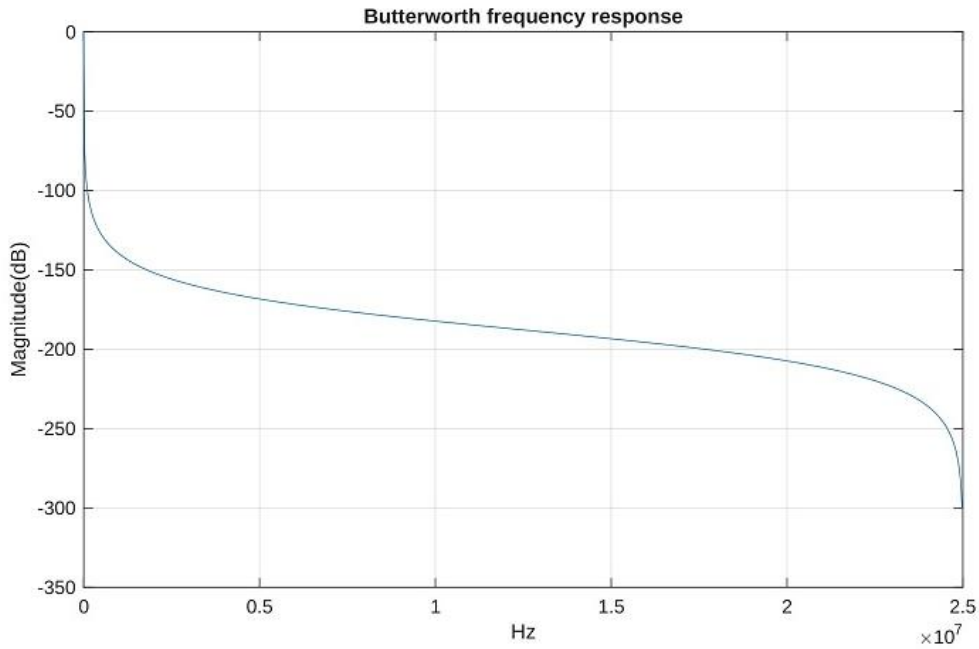
Based on our raw current signal, we are required to design a second-order filter with a cut-off frequency of $f_c = 320$ Hz. Therefore, the calculations are carried out using the following equations.

$w_c = 2\pi f_c = 2\pi \times 320 \approx 2010\, rad/sec$ (4)

H(s) = $\frac{W_c^2}{S^2+\sqrt{2}w_c+w_c^2}$ (5) H(s) = $\frac{4.04\times10^6}{S^2+2843s+4.04\times10^6}$ (6) , $\left|H(_{jw_c})\right| = \frac{1}{\sqrt{2}} \approx -3dB$ (7)

$s_{1,2} = -\frac{w_c}{\sqrt{2}} \pm j\frac{w_c}{\sqrt{2}}$ (8)

$s_{1,2} = -1421 \pm j1421$ (9)



5 Figure: Butterworth Filter Frequency Response

The Butterworth filter is characterized by a maximally flat magnitude response in the passband. This means that, unlike Chebyshev or elliptic filters, it does not exhibit ripples in either the passband or stopband. As a result, all frequency components below the cutoff frequency are passed with nearly equal gain, preserving the original signal shape. At the cutoff frequency, the magnitude of the Butterworth filter drops to −3 dB (approximately 70.7% of the maximum amplitude). Beyond this point, the filter begins to attenuate higher-frequency components smoothly and monotonically. The rate of attenuation depends on the order of the filter, higher order Butterworth filters provide a steeper roll-off, leading to stronger suppression of high frequency noise.

Mathematically, the magnitude response of an n-th order Butterworth low-pass filter is given by:

$$| H(j\omega) | = \frac{1}{\sqrt{1 + \left(\frac{\omega}{\omega_c}\right)^{2n}}}$$

where:

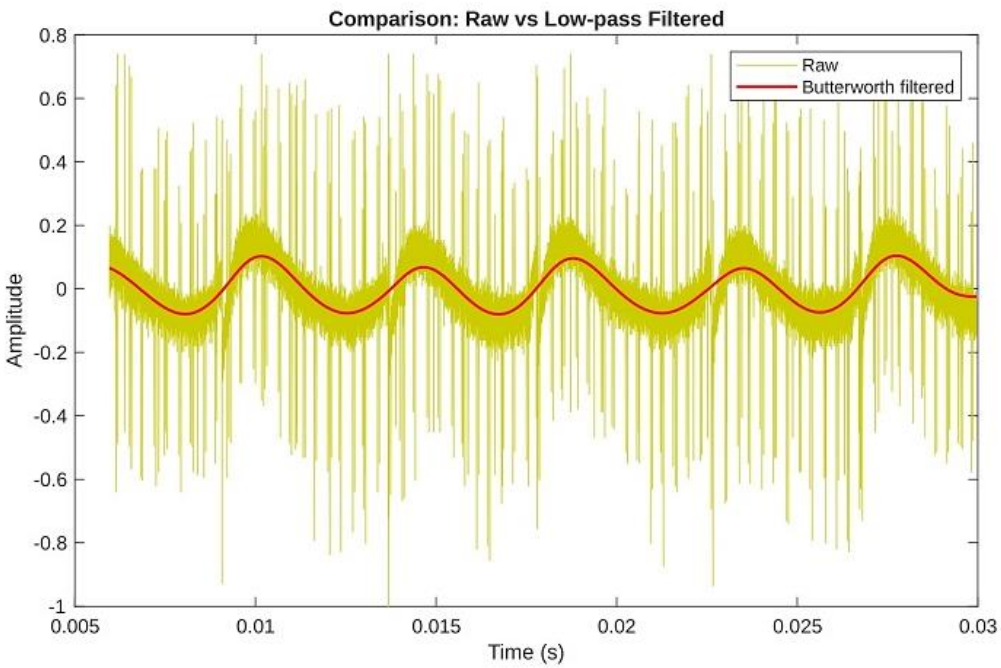
- ω is the angular frequency,
- ω_c is the cutoff angular frequency,
- n is the filter order.

This frequency response ensures a smooth transition from passband to stopband, which minimizes phase and amplitude distortion in the time-domain signal. Consequently, when applied to the raw signal, the Butterworth filter effectively removes high-frequency noise while maintaining the integrity of the fundamental waveform, as observed in the filtered output.

3. Implementing the Butterworth filter

Reduce the effect of noise and distortion present in the measured signal, a Butterworth low-pass filter was applied. The raw signal, shown in the figure 2, contains significant high-frequency components superimposed on the desired waveform. These unwanted components are mainly caused by commutator effects, interference, and system imperfections, which obscure the true behaviour of the signal. The number of spikes per revolution is equal to twice the number of sectors of the commutator. In the case of three-pole motors, six commutation spikes per revolution are generated, three per brush. These spikes are a problem for big DC motors, where the EMI causes a lot of problems, while they are almost negligible for very small motors. The Butterworth filter was selected because of its maximally flat frequency response in the passband. Unlike other filter types that introduce ripples, the Butterworth filter provides a smooth and monotonic response, ensuring that the amplitude of the desired signal components is preserved while attenuating higher-frequency noise. This characteristic makes it particularly suitable for signal smoothing and waveform reconstruction. A low-pass configuration was implemented with a cutoff frequency chosen to retain the fundamental frequency and its relevant harmonics while suppressing high-frequency noise. As a result, frequency components above the cutoff are significantly attenuated without introducing abrupt transitions in the time domain.

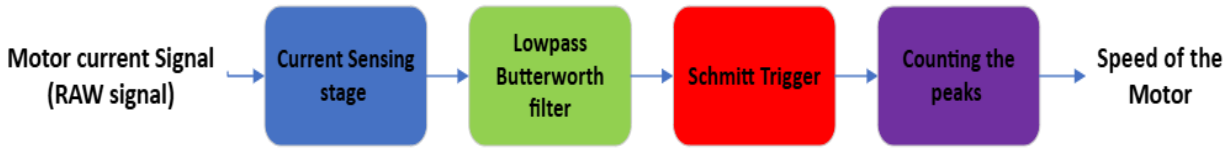
As illustrated in the figure 6, the filtered signal (Butterworth filtered) is noticeably smoother than the raw signal. The rapid fluctuations and high-frequency noise present in the original waveform are greatly reduced, revealing the underlying signal shape more clearly. This improvement enhances signal interpretability and allows for more accurate analysis of amplitude, phase, and timing characteristics.



6 Figure: Comparison of Raw signal vs low pass filtered after applying

4. Ripple to Pulse Conversion

To effectively count and convert current ripple into readable pulses, the current flowing through the BDC motor undergoes various signal conditioning and filtering stages. Chronologically, the stages are: current sensing stage, band-pass filter stage, Schmit trigger stage, and edge counting stage. Figure 7 illustrates the output signals of each ripple conversion stage from the raw current signal up to the digital pulses that will be used to determine the overall speed of DC motor.



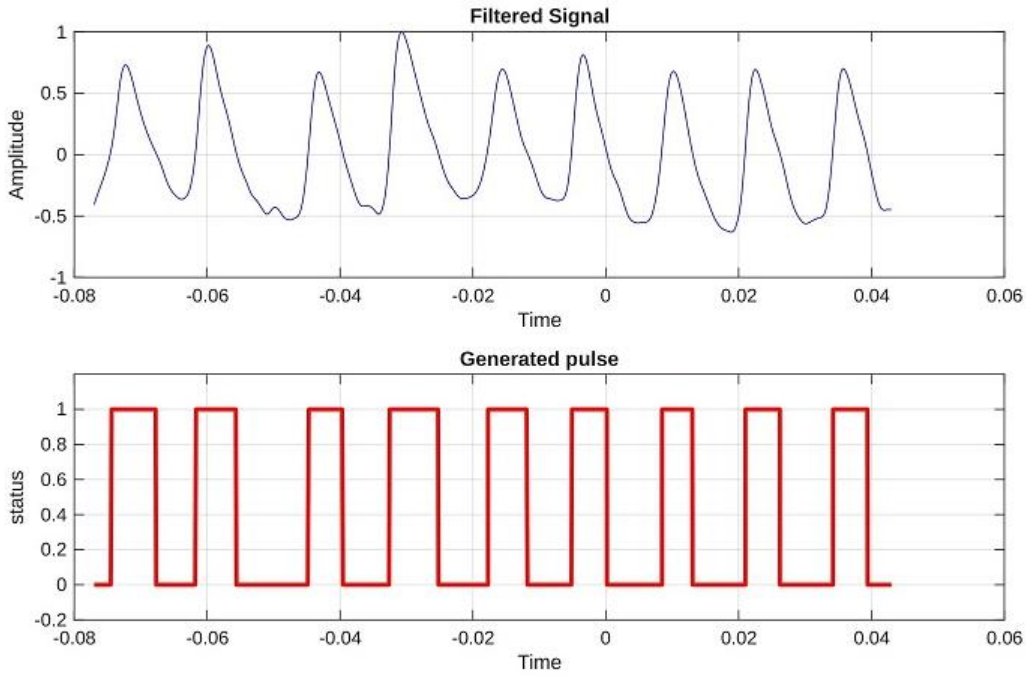
7 Figure: Ripple Conversion Stages Output Signals

The number of brushes and commutator poles are fixed in a specific motor. This is why the number of ripples in a complete rotation is constant. The ripple repeats in every rotation and the number of ripples per revolution can be calculated.

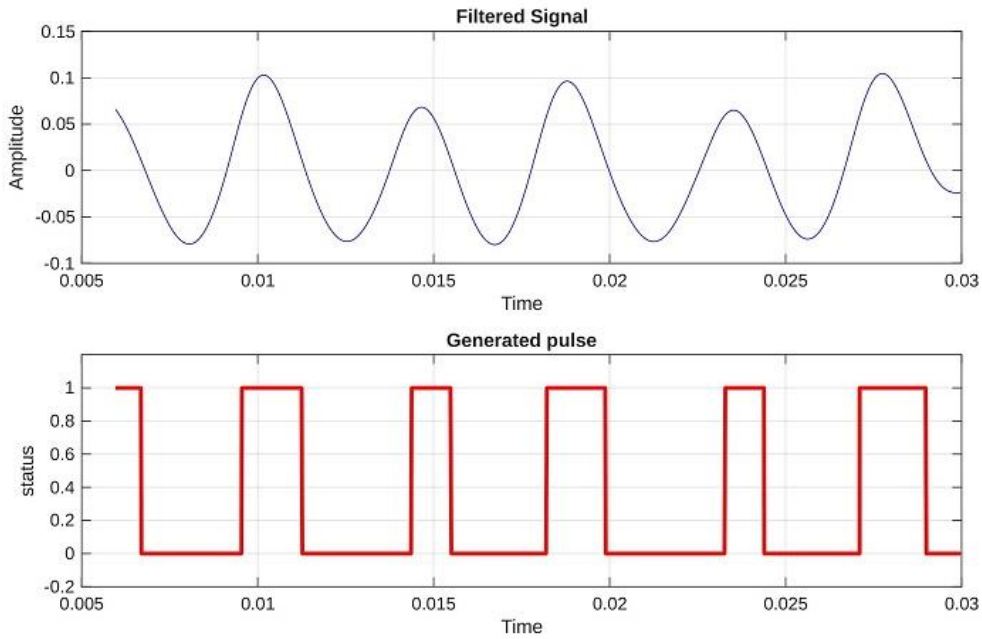
4.1 Pulse Counting Method for DC Motor Frequency Measurement

After the current signal is sensed from the DC motor, it is processed through a Butterworth low-pass filter to remove high-frequency noise and distortions. The filtered signal represents a smooth periodic waveform whose frequency corresponds to the rotational behavior of the motor. To enable accurate frequency measurement, this analog signal is converted into a digital pulse signal using a Schmitt trigger. The Schmitt trigger compares the filtered signal against predefined threshold levels and generates a clean square-wave pulse train. This process eliminates amplitude variations and noise sensitivity around the zero-crossing points, resulting in stable and well-defined pulses, as shown in the generated pulse waveforms.

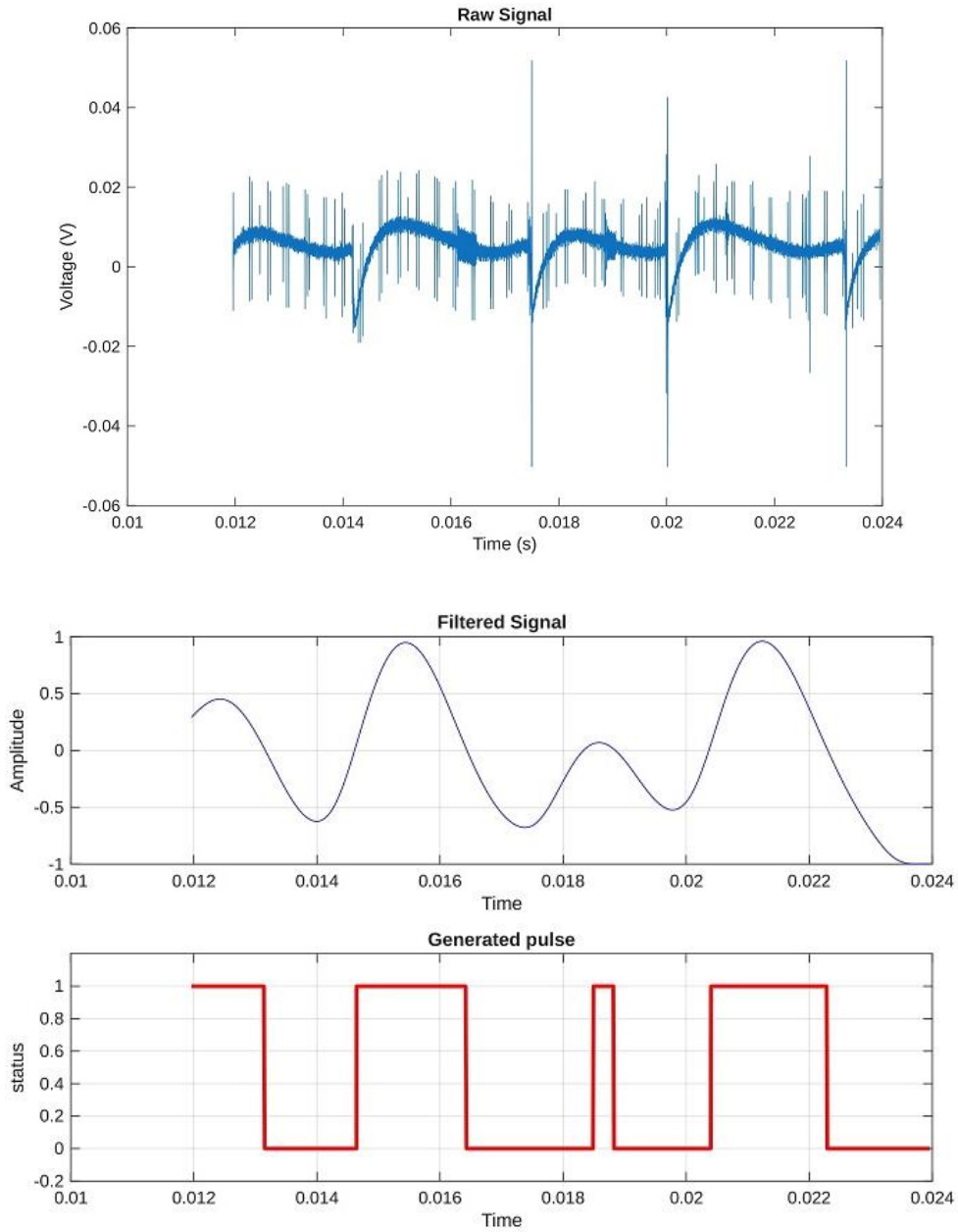
The motor frequency is determined by measuring the time period and counting the number of pulses within a given observation window. Each pulse corresponds to one cycle (or a defined fraction of a cycle) of the filtered signal. The period T of the waveform is calculated as the time interval between two consecutive rising (or falling) edges of the pulse signal. Once the period is known, the frequency $f = \frac{1}{T}$ is obtained from period.



8 *Figure: Filtered signal and pulse generation for a DC motor operating at 20 Hz*



9 *Figure: Filtered signal and pulse generation for a DC motor operating at 75 Hz*



***10** Figure: Raw and Filtered signal and pulse generation for a DC motor operating at 113 Hz*

This pulse-counting approach allows accurate frequency measurement over a wide operating range. As illustrated in the figures, lower motor speeds (around 20 Hz) produce fewer pulses over the same time interval, whereas higher motor speeds (up to 113 Hz) generate a denser pulse train. The increase in pulse repetition rate directly reflects the increase in motor frequency. By relying solely on pulse timing and counting, this method avoids dependency on signal amplitude and the effects of commutator. Consequently, the combination of Butterworth filtering and Schmitt trigger pulse generation provides a reliable and precise technique for estimating the BLDC motor frequency across the range of 20 Hz to 120 Hz.

5. Conclusion and Future work

5.1 summary of result

This application note describes how current ripple signals are successfully converted into countable pulses using a Butterworth low-pass filter followed by a Schmitt trigger. The generated pulses are then counted and processed to estimate the motor speed accurately. Each commutation event corresponds to a single current ripple; therefore, counting the number of ripples over one complete motor revolution enables not only speed estimation but also the determination of the motor's position relative to its initial state. The implementation of ripple counting for brushless DC (BLDC) motor positioning significantly reduces the overall system cost by eliminating the need for additional position sensors. Furthermore, the system incorporates safety features that protect the motor during fault conditions, ensuring reliable and robust operation.

5.2 Further enhancement

Sensorless position and speed control significantly reduces system cost and design complexity by eliminating the need for physical sensors. The proposed algorithm reliably detects motor position and speed across a wide range of applications and operating voltages. The tuning example presented in this document demonstrates the importance and benefits of accurately selected parameters for achieving stable and precise performance. Although certain challenges remain, particularly at very low-speed operation, effective workarounds and signal-processing techniques can be applied to mitigate these limitations. In this work, experimental data were collected using a propeller-driven motor, a current-sensing PCB, an oscilloscope and a Microlab Box, which were used to commission the propeller at desired rotational speed. The acquired signals were subsequently processed and analyzed using MATLAB software.

As future work, the proposed algorithm can be implemented on a Micro Lab Box to enable real-time signal processing and simultaneous visualization of waveforms during DC motor operation. This implementation would further validate the approach under practical operating conditions and support real-time monitoring and control.

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